



Machine learning in the early detection of endometriosis: a literature review on symptom clustering and imaging integration

A literature review

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Received: June 22, 2025

Revised: September 1, 2025

Accepted: September 6, 2025

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ABSTRACT

Endometriosis is a gynecologic inflammatory condition that affects up to 10% of reproductive-aged women worldwide. The disease exhibits heterogeneous presentations and is associated with a prolonged diagnostic delay, often exceeding seven years, because existing diagnostic modalities such as transvaginal ultrasound, magnetic resonance imaging, and the biomarker cancer antigen 125 (CA-125) are suboptimal. This review examines how machine learning (ML) is playing an increasingly significant role in early, non-surgical endometriosis diagnosis through two main approaches: symptom clustering and imaging integration. Unsupervised ML algorithms such as k-means, partitioning around medoids, and Bayesian networks have demonstrated success in identifying clinically informative endometriosis phenotypes from patient-reported symptoms and electronic health records. Concurrently, ML models such as convolutional neural networks and radiomics approaches have achieved high accuracy in lesion detection from imaging data, in some cases surpassing human interpretation. Despite these advances, significant challenges remain, including limited access to large, annotated multimodal datasets, the absence of widely accepted evaluation standards, and concerns regarding interpretability and generalizability. Multicenter, integrative studies and the incorporation of explainability techniques are recommended as potential strategies to address these gaps. Finally, multimodal ML approaches that combine symptomatology and imaging data hold substantial promise for reducing diagnostic delays, facilitating early intervention, and improving clinical outcomes in the management of endometriosis.

Keywords: Artificial intelligence; Endometriosis; Machine learning

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INTRODUCTION

Endometriosis is a chronic, inflammatory gynecological disorder characterized by the presence of endometrial-like tissue outside the uterus, which leads to pain, infertility, and reduced quality of life. It affects approximately 10% of reproductive-aged women worldwide, with some estimates suggesting an even higher prevalence due to underdiagnosis [1]. The most common symptoms include dysmenorrhea (painful menstruation), chronic pelvic pain, dyspareunia (pain during intercourse), and subfertility, although symptom severity does not always correlate with disease stage [2]. In addition to physical suffering, endometriosis significantly affects mental health, work productivity, and social relationships, with many patients reporting anxiety, depression, and reduced overall well-being [3]. The economic burden is also considerable, with healthcare expenditures and productivity losses amounting to billions of dollars annually [4]. Given its widespread impact, enhancing diagnostic and therapeutic strategies is critical for mitigating the long-term consequences of this debilitating condition.

Despite its prevalence, endometriosis remains notoriously difficult to diagnose, with an average delay of 7 to 10 years from symptom onset to definitive diagnosis [5]. The gold standard for diagnosis is laparoscopic surgery with histological confirmation, an invasive procedure that carries risks such as infection, bleeding, and complications related to anesthesia [6]. Noninvasive imaging techniques, including transvaginal ultrasound (TVUS) and magnetic resonance imaging (MRI), are frequently employed but demonstrate variable sensitivity and specificity depending on lesion type and operator expertise [7]. Furthermore, biomarkers such as cancer antigen 125 (CA-125) lack sufficient accuracy for use as stand-alone diagnostic tools, leading to frequent misdiagnoses with conditions such as irritable bowel syndrome or pelvic inflammatory disease [8]. These diagnostic limitations underscore the urgent need for more reliable and less invasive methods to reduce delays and improve patient outcomes. The protracted diagnostic journey for endometriosis patients highlights the importance of developing early, noninvasive diagnostic tools. Early detection could prevent disease progression, reduce chronic pain, and preserve fertility by enabling timely intervention [9]. Emerging research is investigating liquid biopsies, proteomics, and microRNA profiling as potential diagnostic strategies, although none have yet achieved widespread clinical adoption [10]. In addition, integrating patient-reported symptoms with imaging findings may enhance

diagnostic accuracy without immediate reliance on surgery [11]. Given the shortcomings of current methods, there is a pressing demand for innovative approaches that combine clinical data with advanced analytical techniques to streamline diagnosis and improve patient care.

Artificial intelligence (AI) and machine learning (ML) are transforming healthcare by enabling automated pattern recognition, predictive modeling, and decision-support systems. In medical imaging, AI algorithms have demonstrated considerable success in detecting malignancies, cardiovascular abnormalities, and neurological disorders [12]. These technologies excel in processing large datasets, identifying subtle patterns, and enhancing diagnostic precision capabilities that could help address key challenges in endometriosis diagnosis [13]. For example, ML models trained on ultrasound or MRI datasets can improve lesion detection, while natural language processing can extract diagnostic information from electronic health records (EHRs) [14]. As AI continues to evolve, its application in endometriosis holds promise for reducing diagnostic delays and improving classification accuracy.

This review aims to critically examine the current diagnostic challenges in endometriosis and explore the potential of AI-driven solutions to overcome these barriers. We evaluate the limitations of existing diagnostic tools, analyze recent developments in biomarker research and imaging technologies, and discuss how AI can integrate multimodal data to improve accuracy. Furthermore, we highlight ongoing research, clinical applications, and future directions for AI in endometriosis care. By synthesizing the latest evidence, this review seeks to inform clinicians, researchers, and policymakers about the transformative potential of AI in addressing one of gynecology's most persistent diagnostic challenges.

MATERIALS AND METHODS

Electronic databases, including PubMed/MEDLINE, Web of Science, and Google Scholar, were searched for case reports, case series, case-control studies, cohort studies, cross-sectional studies, and review articles from each database's inception to May 2025. The search terms included 'Endometriosis,' 'Machine learning,' 'Symptom clustering,' and 'Artificial intelligence,' among others. Studies were included if they investigated the application of AI or ML to the diagnosis or early detection of endometriosis, reported original data (including case-control, cohort, cross-sectional studies, or randomized studies) or systematic/narrative reviews directly relevant to ML in endometriosis, used symptom-based, imaging-based,

or multimodal datasets; and were published in English. Both prospective and retrospective studies were eligible, and no restrictions were placed on patient age or geographic setting. The majority of the studies included were small to medium-sized, often retrospective single-center registries with less than 100 up to several hundred patients. The patient populations were enrolled predominantly from the gynecology specialty clinics and may have had limited generalizability to more prevalent community practice. In particular, most studies defined endometriosis by surgical/pathologic diagnosis, the current gold standard for diagnosis, while others used clinical diagnosis or imaging study findings such as ultrasound or MRI. This was conducted as a narrative review rather than a systematic review; therefore, no formal quality assessment or strict exclusion criteria were applied. Study inclusion was primarily based on relevance to the research question and the authors' judgment of contribution to the evolving field.

DIAGNOSTIC CHALLENGES IN ENDOMETRIOSIS

Heterogeneity of symptoms

One of the primary obstacles in diagnosing endometriosis is the wide variability in clinical presentation. While some patients experience severe pelvic pain and infertility, others remain asymptomatic, with lesions discovered incidentally during surgery for unrelated conditions [15]. Symptom patterns also differ by lesion location; for instance, deep infiltrating endometriosis (DIE) may cause bowel or urinary symptoms, whereas ovarian endometriomas are more commonly associated with dysmenorrhea [16]. This heterogeneity complicates diagnosis, as clinicians must distinguish endometriosis from other conditions with overlapping symptoms, such as adenomyosis, interstitial cystitis, and gastrointestinal (GI) disorders [17]. The absence of a definitive symptom profile necessitates a multidisciplinary diagnostic approach, further delaying identification and treatment.

Average diagnostic delay (typically 7 to 10 years)

The prolonged diagnostic delay in endometriosis typically ranging from 7 to 10 years reflects systemic shortcomings in awareness, clinical evaluation, and diagnostic practices [18]. Studies show that adolescents and young women often experience even longer delays, as their symptoms are frequently dismissed as 'normal' menstrual pain [18,19]. Contributing factors include clinician bias, normalization of pelvic pain,

and limited access to specialized care [20]. These delays permit disease progression, thereby increasing the risk of chronic pain, infertility, and surgical complications [21]. Reducing this delay requires enhanced clinician education, standardized symptom-assessment tools, and improved diagnostic technologies to support early intervention.

Limitations of current diagnostic tools (ultrasound, MRI, and CA-125)

Although TVUS and MRI are widely used for endometriosis detection, their diagnostic accuracy depends on lesion type, operator expertise, and imaging protocols [22]. TVUS is effective for detecting ovarian endometriomas but is less sensitive for superficial peritoneal or deep infiltrating lesions [23]. MRI provides superior soft-tissue contrast but is costly and not universally accessible [24]. Serum biomarkers such as CA-125 are often elevated in endometriosis but lack specificity, as they may also increase in ovarian cancer, pregnancy, and other inflammatory conditions [25]. These limitations underscore the need for more reliable and cost-effective diagnostic tools that can detect endometriosis across its diverse manifestations without invasive procedures.

Importance of integrating symptomatology and imaging

Given the limitations of individual diagnostic methods, a multimodal approach that combines symptom assessment, imaging, and biomarker analysis may improve diagnostic accuracy. Recent studies suggest that structured symptom questionnaires (e.g., the Endometriosis Symptom Diary) can aid in stratifying patients for further evaluation [26]. When paired with AI-enhanced imaging analysis, these tools could facilitate earlier identification of high-risk individuals [27]. For instance, ML algorithms can process ultrasound or MRI data to detect subtle lesions that may be overlooked by human evaluators [28]. Integrating patient history with AI-driven imaging interpretation has the potential to reduce reliance on laparoscopy, thereby enabling faster and less invasive diagnoses. Future research should prioritize validating such integrated models in diverse clinical settings to ensure broad applicability.

MACHINE LEARNING IN MEDICINE: A PRIMER

ML is a subfield of AI that encompasses computational methods enabling computers to learn from medical records and prior experiences to categorize or predict outcomes without

explicit programming [29,30]. The three primary types of ML are supervised learning, unsupervised learning, and deep learning. Deep learning is a more advanced approach that builds upon the first two paradigms [31,32]. To predict outcomes for new, unseen data, supervised learning requires algorithms trained on labeled datasets, in which the correct outputs are already known. This method is frequently applied in diagnostic contexts for tasks such as regression and classification, where prior patient data are used to train models to differentiate between diseased and healthy states [31,33]. Unsupervised learning, by contrast, does not utilize labeled data but instead identifies hidden patterns or clusters. This approach is particularly valuable for uncovering phenotypes or disease subtypes that may not be immediately apparent to clinicians [31,34]. Deep learning, also referred to as deep neural network learning, is especially powerful for analyzing medical images and signals, employing multilayered neural networks to automatically extract features and recognize complex patterns [35,36].

ML is increasingly applied in clinical settings because it can identify patterns within large and complex medical datasets. ML algorithms can assist clinicians in developing more effective treatment strategies and making more accurate diagnoses by analyzing patient information. For example, an ML model can use hospital data on symptoms and treatment outcomes to predict which therapies are most likely to be effective for specific patient groups [35,37]. These algorithms, often referred to as classifiers, can accurately distinguish between diagnostic categories depending on the model architecture and the quality of the data [37,38]. ML is also used for risk prediction, employing algorithms such as support vector machines (SVMs) and convolutional neural networks (CNNs) to identify high-risk patients at earlier stages [31,38]. When applied collectively, these approaches enable clinicians to detect diseases sooner, design personalized treatment plans, and achieve improved outcomes across diverse medical fields.

One area where ML has demonstrated particular value is gynecology, where it enhances the accuracy and efficiency of disease detection, risk assessment, and clinical decision-making, especially in cancer diagnosis and pregnancy-related complications [39,40]. In cancer care, ML and deep learning models are applied to diagnose cervical, ovarian, and endometrial cancers through image analysis, biomarker discovery, and individualized risk prediction [41]. These models frequently outperform conventional diagnostic methods in sensitivity and accuracy, thereby facilitating earlier detection and reducing treatment delays [41-43].

Beyond oncology, ML also plays an important role in predicting and managing a wide range of gynecological conditions. Algorithms such as decision tree ensembles and naïve Bayes classifiers have demonstrated high effectiveness in predicting conditions including postpartum depression, anemia, preeclampsia, gestational diabetes, and miscarriage achieving accuracies of up to 86% using symptom and clinical data [44-46]. In maternal-fetal medicine, ML facilitates the early identification of complications such as preterm birth, cesarean delivery, and perinatal death, often through the integration of clinical metrics with imaging data [40,44]. Furthermore, AI and ML are transforming gynecologic imaging by automating image classification, assisting radiologists in their diagnostic tasks, and promoting greater consistency and objectivity in clinical decision-making [42,43]. These developments illustrate how the fundamental principles of ML are being applied to integrate clinical care with computational methods, thereby achieving tangible improvements in women's health.

SYMPTOM CLUSTERING IN ENDOMETRIOSIS USING MACHINE LEARNING

ML-based symptom clustering has emerged as a viable strategy for characterizing the complex clinical manifestations of endometriosis. Infertility, pelvic pain, dysmenorrhea, dyspareunia, and GI symptoms are frequently included as key variables for clustering and prediction in endometriosis research (Table 1). While GI symptoms are often misattributed, contributing to diagnostic delays, pelvic pain and dysmenorrhea are particularly prevalent in both adolescent and adult populations. These symptoms are central to patient-reported outcomes as well as clinical data analyses [47-50].

Building on this symptom-based framework, unsupervised ML techniques such as multivariate mixture models, partitioning around medoids (PAM), and k-means clustering have been employed to identify subtypes within endometriosis populations. K-means clustering, for example, has been applied to classify women according to quality-of-life parameters, revealing discrete groups with either high or low quality of life [51]. Similarly, analyses of adolescent EHR data using PAM and mixture models have identified 'classic' (pelvic pain, dysmenorrhea, chronic pain), 'GI' (gastrointestinal-dominant), and 'feature-absent' phenotypes [49]. In addition, anatomical pain sites have been mapped using Bayesian network analysis, which has also been applied to identify patterns in pain symptomatology [47]. Furthermore, mixed-membership

Table 1. Summary of symptom clustering in endometriosis using machine learning

Key aspect	Methods used	Data source	Main findings/insights
Key symptoms	Pelvic pain, dysmenorrhea, dyspareunia, infertility, GI issues	Patient-reported, EHRs	Central to clustering and prediction models [47-51]
Unsupervised learning	k-means, PAM, mixture models, Bayesian networks	Surveys, EHRs, self-tracking	Identified classic, GI, and feature-absent phenotypes; robust to data variability [47,49-51]
Patient-reported/EHR data	Surveys, self-tracking, EHR notes	Large cohorts, clinical notes	Enabled detection of symptom heterogeneity and less-recognized phenotypes [49,51]
Distinguishing disorders	Clustering of symptom constellations	Bayesian networks, clustering	Classic symptom clusters increase endometriosis risk; models help differentiate from other pain disorders [47,49]

GI, gastrointestinal; EHR, electronic health record; PAM, partitioning around medoids.

models have shown promise in handling multimodal, self-tracked symptom data, thereby improving the clinical relevance of identified subtypes [50].

Increasingly, studies incorporate EHR data and patient-reported outcomes obtained from large-scale surveys and symptom-tracking applications to support these computational approaches and capture the full heterogeneity of endometriosis. These data sources enable researchers to identify phenotypes directly from real-world experiences through unsupervised learning [50,51]. Clinical documentation has also been analyzed with EHR-based clustering, which has revealed a spectrum of symptom profiles and their associations with treatment patterns [49].

Crucially, these clustering models not only help define endometriosis subtypes but also aid in distinguishing the condition from other pelvic pain disorders. ML techniques have demonstrated higher predictive value for diagnosing endometriosis compared with other pain syndromes by identifying specific symptom constellations, such as the co-occurrence of chronic pelvic pain, subfertility, and dyspareunia [47]. Furthermore, studies indicate that women with classic phenotypes typically receive more hormonal or pain-related interventions, whereas women with non-classic or GI-dominant presentations may receive less clinical attention [49,50]. These findings underscore the clinical value of detailed symptom profiling in improving differential diagnosis and guiding individualized therapeutic approaches.

IMAGING-BASED MACHINE LEARNING MODELS

TVUS and MRI are among the most widely used noninvasive methods for detecting endometriosis, with each offering dis-

tinct advantages and disadvantages. TVUS is a first-line imaging modality for endometriosis because of its minimal risk, low cost, and high diagnostic performance, with reported sensitivity ranging from 71% to 98% and specificity from 92% to 100%. TVUS can detect endometriotic lesions in the uterine lining as small as 5 mm, typically appearing as hypoechoic masses, while peritoneal lesions larger than 5 mm are classified as DIE. The ‘tenderness-guided’ technique and the ‘sliding sign’ are commonly used to assess lesion depth and the presence of endometriosis in the pouch of Douglas (POD) [52].

MRI is considered a second-line imaging modality following TVUS due to its higher cost and limited availability. However, MRI offers important advantages, including the ability to evaluate larger pelvic structures such as the bowel, ureters, and extra-pelvic lesions, as well as high sensitivity for fibrin degradation products that facilitate the detection of obscured lesions [52]. Although TVUS and MRI remain essential in endometriosis diagnosis, image interpretation is challenging. In this context, ML models such as CNNs and radiomics approaches have shown potential for improving diagnostic accuracy.

CNNs and radiomics as tools in medical imaging

Diagnosing endometriosis using noninvasive methods such as TVUS and MRI remains challenging, with inconsistencies even in expert interpretation. However, the emergence of AI-driven, imaging-based ML models such as CNNs and radiomics offers the potential for more accurate and timely diagnosis when trained on large datasets of annotated images that capture subtle patterns of endometriosis often imperceptible to the human eye [53]. CNNs are a class of deep learning models capable of processing vast amounts of data; they are specifically designed for tasks such as image pattern

recognition. In medical contexts, CNNs are frequently applied for detection, classification, and segmentation. Among CNN-based models, U-Net is particularly well suited for medical image segmentation. This 'U-shaped' neural network has been extensively used in research on gynecological and oncological conditions. Notably, a 2024 study demonstrated that U-Net achieved highly accurate segmentation of endometriotic lesions, with a Dice coefficient of 0.977 suggesting that CNN-based models trained on sufficiently large datasets can substantially improve the accuracy of noninvasive endometriosis diagnosis [53].

In addition to CNNs, another imaging-based AI technique is radiomics, a process that extracts and analyzes quantitative features from medical images and transforms them into high-dimensional data such as shape, texture, and intensity. The radiomics workflow typically involves several key steps: image acquisition, preprocessing, segmentation, feature extraction, and model building [54,55]. Radiomics can detect subtle patterns and variations in endometriosis that are often overlooked during traditional interpretation, thereby providing a more comprehensive diagnostic approach [54,55]. For example, in a study that employed radiomics to develop a model distinguishing ovarian endometriomas from dermoid cysts, the model achieved an area under the curve (AUC) of 0.981 and an accuracy of approximately 94% [56]. A recent advancement in this field is the Human-AI Collaborative Multimodal Multirater Learning (HAICOMM) method, which integrates radiomics with collaborative human-AI models to further enhance diagnostic performance in endometriosis. HAICOMM demonstrated significant improvements in the classification of POD obliteration on MRI images, outperforming interpretations made by either clinicians or AI models alone [57]. Together, CNNs and radiomics exemplify how AI holds substantial promise for improving endometriosis diagnosis through noninvasive methods, thereby reducing diagnostic delays and improving patient outcomes.

ML models in detecting endometrioma, DIE, and peritoneal lesions

Superficial peritoneal endometriosis

In one study, several ML algorithms including random forest and gradient boosting were trained on data from clinical history and physical examinations in women with chronic pelvic pain but no detectable abnormalities on imaging. The best-performing models achieved a sensitivity of 79.3% and a specificity of 74.2% for predicting superficial peritoneal endometriosis (SPE), with subsequent surgical confirmation. Key pre-

dictive features included oligomenorrhea, bladder pain syndrome, and irritable bowel syndrome [58]. Additionally, another study reported that a random forest model using a saliva microRNA signature (89 miRNAs) achieved 100% sensitivity, specificity, and AUC for detecting the SPE phenotype. This finding suggests a highly accurate and noninvasive diagnostic tool that could potentially replace invasive procedures such as laparoscopy [59].

DIE and endometrioma

In a study on DIE, the inclusion of 16 clinical symptoms in an ML algorithm yielded AUC values ranging from 0.91 to 0.95 in training datasets and from 0.66 to 0.92 in test datasets. These models may serve as screening tools for DIE in both gynecological and general practice settings [28]. For endometriomas, differentiation from SPE and DIE relies on imaging modalities such as ultrasound and MRI, as well as distinct cytokine signatures in peritoneal fluid. Future ML models incorporating these biomarkers have the potential to further improve diagnostic accuracy [60].

Human vs. ML diagnostic performance

Although ML models have shown promise in the medical imaging of endometriosis, a key question remains: do ML models outperform clinicians in diagnosis? Comparative evaluation of human and ML diagnostic performance is still an emerging area of research. ML approaches aim to improve accuracy, speed, and noninvasiveness, whereas current clinical standards continue to rely heavily on clinical evaluation and invasive procedures such as laparoscopy.

On one hand, several studies support the diagnostic value of ML models using noninvasive methods. For instance, a 2023 study employing a CNN-based deep learning model, Visual Geometry Group (VGG-16), applied to MRI images reported a sensitivity of 84.15% and a specificity of 83.86%, indicating that ML has the potential to reduce false negatives and diagnostic delays [61]. Other studies applying ML algorithms to detect ultrasonographic signs reported sensitivities and specificities between 70% and 80%. Well-known models such as random forest and extra-trees achieved an AUC of 0.76, suggesting moderate to good diagnostic performance [62]. On the other hand, a systematic review comparing TVUS and MRI reported sensitivity and specificity values across different lesion locations (Table 2). TVUS demonstrated similar sensitivity to MRI but slightly higher specificity. These findings suggest that both methods are reliable, though performance varies depending on lesion location [22].

Table 2. Comparison between sensitivity and specificity of magnetic resonance imaging in various locations

Location	Sensitivity (%)	Specificity (%)
Recto-sigmoid	85	95
Recto-vaginal septum	66	97
Uterus	70	93

Limitations of imaging-based ML

Although ML techniques have shown promising results in diagnostic accuracy, traditional methods remain important in clinical practice. The current standard for diagnosing endometriosis involves clinical evaluation, imaging modalities such as MRI and ultrasound, and confirmatory laparoscopy. This reliance often results in diagnostic delays due to the complex and heterogeneous nature of symptoms. A major limitation of ML is its tendency to overfit, particularly when trained on small datasets. Consequently, researchers emphasize the need for large, well-curated datasets to improve training and to account for the variability of symptoms experienced throughout a patient's lifetime. Addressing this challenge requires a comprehensive strategy that integrates ML with imaging and additional diagnostic approaches, such as biomarker testing, to more effectively detect endometriosis [63]. Despite encouraging findings, the field still requires systematic protocols for image preprocessing and feature extraction across imaging modalities, as well as clinical validation through large-scale cohort studies to ensure reliability and generalizability. Collectively, current evidence suggests that human interpretation remains a necessary component of endometriosis diagnosis, despite the limitations faced by clinicians. Only once modern AI technologies are consistently validated and shown to outperform conventional methods can they be considered a standalone alternative.

INTEGRATION OF SYMPTOM CLUSTERING WITH IMAGING DATA

Although ML models such as CNNs and radiomics have shown substantial potential in detecting endometriotic lesions, their accuracy is limited when the heterogeneous clinical presentations of patients are not considered. To address this limitation, recent studies have focused on integrating symptom clustering with advanced ML imaging data. By jointly evaluating patient-reported symptoms and radiological findings, ML models can provide a more holistic and individualized approach to diagnosing endometriosis. This section

explores ML methodologies that emphasize such integrative strategies in endometriosis research [64]. Multimodal ML refers to the use of algorithms that analyze multiple types of data. In the context of endometriosis, this involves combining clinical symptoms with imaging data to improve diagnostic outcomes. The application of multimodal ML is particularly relevant for complex and heterogeneous diseases such as endometriosis. Recent advances highlight the importance of combining clinical, symptom, and imaging data using multimodal ML to achieve more accurate and comprehensive diagnostic solutions. For instance, in one study that developed an ML model incorporating both imaging data and patient-reported symptoms, the results demonstrated high sensitivity and specificity in diagnosing endometriosis [65].

Fusion strategies, benefits, and challenges in multimodal ML

A systematic review of fusion techniques found that early fusion in which raw data from different modalities are combined before training the model is the most commonly used strategy compared with other fusion methods. This approach benefits the model by enabling it to learn cross-modal representations from the outset [66]. Both conventional ML models (e.g., decision trees and SVMs) and deep learning models (e.g., CNNs and transformers) can be integrated into multimodal systems. Additionally, Bayesian networks have been applied to cluster pain symptoms and relate them to surgical diagnoses, thereby improving identification of symptom patterns in endometriosis [67]. Ensemble and decision tree methods are also frequently employed to combine clinical and imaging data, making them useful in the broader context of leveraging multimodal ML for disease diagnosis [47].

The benefits of multimodal ML include improved diagnostic accuracy through the integration of symptom patterns, imaging features (e.g., MRI), and clinical evaluations, which helps overcome the limitations of standalone clinician assessments or ML models [16]. Multimodal ML can also facilitate the identification of distinct subtypes of endometriosis, thereby supporting earlier and more personalized treatment. Furthermore, ML models that integrate imaging with clinical and symptom data can achieve more accurate mapping of lesion location and depth, which is critical for surgical planning [51]. Nevertheless, several challenges remain. Variability in symptom reporting and imaging protocols may complicate the data processed by ML models. Many studies are also constrained by small or single-center cohorts, limiting generalizability and statistical significance. Finally, the complexity of advanced ML

models such as ensembles can make them difficult for clinicians to interpret and trust in decision-making [68].

The integration of multimodal ML has the potential to advance current diagnostic guidelines for endometriosis, but addressing existing challenges and utilizing large datasets will be essential for widespread clinical adoption.

LIMITATIONS AND CURRENT GAPS IN THE LITERATURE

Although advances in technology applied to AI- and ML-based approaches can significantly reduce diagnostic delay and increase sensitivity of noninvasive endometriosis diagnosis, downstream clinical impact is complex. Treatment choice continues to remain symptomatic and empirical in practice today with excessive use of symptom suppression by hormones, analgesics, or surgery, rather than disease-phenotype-guided tailored approaches [2,9,21]. As has been emphasized by recent criticism, endometriosis continues to be managed largely as chronic pain syndrome, symptom-load driven treatment rather than overt diagnostic subtyping [16]. Thus, even if ML models enhance the likelihood of earlier or more accurate detection, their short-term impact on therapeutic gain may be blunted unless diagnoses progress are integrated into evidence-based treatment pathways. Follow-up analyses need to move beyond initial trend toward diagnosis itself to ascertain if early diagnosis with AI is being translated into better clinical results, such as reduced pain, preservation of fertility, better quality of life, or more personalized operative planning. Without this translation step, better diagnostic equipment is technology advancement with no-holds-barred therapeutic correlations. Despite the promising advances in applying ML to the early diagnosis of endometriosis, several limitations and gaps remain in the current literature. One of the most significant limitations is the lack of large, labeled datasets that combine clinical symptoms with imaging data, which hinders the development and validation of robust predictive models. Most recent studies have focused exclusively on either symptom clustering or imaging interpretation, rather than leveraging the synergistic benefits of multimodal data integration. For example, Wang et al. [57] introduced the HAICOMM model to address this gap by combining T1- and T2-weighted MRI images with domain-annotator labels, but such efforts remain scarce in the literature. Generalizability of existing models is further limited by reliance on single-institution retrospective datasets, which may not adequately represent heterogeneous patient populations. Maleki et al. [69]

emphasized that methodological shortcomings such as data leakage and batch effects can lead to overestimation of model performance and reduce applicability across different clinical settings. In addition, the lack of standardized evaluation metrics and benchmarking protocols hinders meaningful comparison of model performance across studies [69]. Salathuddin et al. [70] advocated for uniform evaluation frameworks to ensure both interpretability and reliability of deep learning models in medical imaging. Interpretability itself remains a major challenge for complex models such as deep learning networks, with limited efforts directed toward incorporating explainability tools that could foster clinician trust and usability [70]. Solutions such as InterpretML (Microsoft Research) have been proposed, but their application to endometriosis research is only beginning to emerge. Addressing these challenges will be critical to transitioning ML-based tools from experimental applications to routine clinical practice in endometriosis diagnosis [71].

CONCLUSION

ML represents a transformative approach to advancing early diagnosis of endometriosis by integrating symptom clustering with imaging data. Current models show strong potential; however, their translation into clinical practice is significantly limited by challenges in data availability, interpretability, and standardization. To encourage widespread clinical adoption at an accelerated pace, studies must continue to tackle; creating multicenter, multimodal data consortia with wide and heterogeneous datasets, standardized benchmarking platforms and assessment measures to make studies comparable, incorporating explainability tools to establish clinician trust and acceptance into routine operating practice, conducting prospective external validation studies to determine real-world generalizability and conducting cost-effectiveness analyses to inform policy planning and the use of healthcare resources. By meeting these priorities, ML-based diagnostic tools can move on from proof-of-concept to clinically viable solutions and ultimately reduce diagnostic delays and improved outcomes for endometriosis in women.

CONFLICTS OF INTEREST

No potential conflict of interest relevant to this article was reported.

ACKNOWLEDGMENTS

The authors would like to thank Dr Noha Farag for her valuable time and contribution.

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Acquisition, analysis, or interpretation of data: ZMR, SMM, SSK, SK, NSF.

Drafting the work or revising: ZMR, SMM, SSK, SK, NSF.

Final approval of the manuscript: ZMR, SMM, SSK, SK, NSF.

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